

# AHMED SOLIMAN

## Software Engineer / Architect

U.S. Citizen | New York City Metropolitan Area | +13477554426 | [Linkedin](#) | [ahmedsoliman.com](#)

## PROFESSIONAL SUMMARY

Software Engineer and Architect at the intersection of Enterprise Data Strategy and Generative AI, with 6+ years designing and leading enterprise-grade distributed data platforms in financial technology. Currently leading the enterprise-wide Storm-to-Snowflake migration at BNY and architecting the development framework for the Eliza Mesh data ingestion platform. Deep expertise in Apache Spark orchestration, Azure Kubernetes Service (AKS) multi-cluster deployments, and end-to-end CI/CD automation. Proven track record designing real-time streaming and batch ingestion systems processing high-volume financial datasets. Skilled in building production-grade AI agents leveraging retrieval-augmented generation (RAG) and LLM-driven workflow automation. Focused on technical leadership — architecting solutions that are compliant, scalable, and built for long-term operational efficiency.

## PROFESSIONAL EXPERIENCE

### BNY (Bank of New York Mellon) — Software Engineer / Full Stack Engineer

New York City Metropolitan Area | 2022 – Present

#### Platform Engineering & Data Processing

- Lead the enterprise-wide Storm-to-Snowflake migration — a large-scale platform modernization encompassing Apache Storm decommissioning, batch workload migration to Apache Spark, and transition toward real-time streaming architectures
- Architect the development framework underpinning both Data Lake and non-lake tenants for Eliza Mesh, enabling scalable, resilient, and governed data ingestion capabilities across the organization
- Designed and implemented core components of a Global Data Platform including the Spark processing engine, Vertica analytical loader, event-driven trigger service, and data ingestion framework REST APIs — from early design through production deployment
- Developed and tuned SQL workflows across PostgreSQL and Vertica to support high-volume ingestion, metadata management, and analytics-oriented processing
- Operated distributed Spark workloads across multi-cluster AKS environments spanning development, QA, and production with high availability and fault tolerance
- Built and maintained Elasticsearch indices for file tracking, event metadata, and data certification; wrote Painless scripts for bulk metadata transformations across the platform
- Led the convergence initiative between two major data platforms, aligning architecture, deployment processes, and operational practices to reduce redundancy and improve maintainability
- Upgraded core platform components to Java 17 and managed Apache Iceberg table maintenance and Nessie catalog operations to ensure data lake reliability

#### DevOps & CI/CD Automation

- Constructed a unified GitLab CI/CD pipeline for all platform components with integrated change management, multi-project triggers, and structured Maven release versioning
- Automated Docker image build and publish workflows across a private container registry, managing deployment artifacts for 6+ distributed services including Storm, ZooKeeper, and Spark Operator
- Managed Helm chart releases for Kubernetes-based microservices and configured RBAC policies for secure Spark job submission across namespaces
- Implemented end-to-end deployment automation for legacy and modern components including artifact download, image build, verification, and cloud-native deployment
- Integrated automated security vulnerability scanning and SonarQube quality gates into GitLab CI/CD pipelines, reducing vulnerability backlog and strengthening the organization's security posture

#### Risk Management & Production Support

- Developed and executed compliant backout plans with rollback procedures for Kubernetes deployments and cloud-hosted services
- Diagnosed and resolved production issues using Spark driver/executor logs, Splunk queries, and Elasticsearch analytics across platform indices
- Proactively remediated platform vulnerabilities in critical services by patching, rebuilding, and redeploying containerized images
- Managed Kubernetes secrets, Spark UI debugging, and StatefulSet/Pod lifecycle operations across production clusters
- Supported SailPoint Certification reporting, SoD violation remediation, and metadata rebranding automation

#### Innovation & AI Engineering

- Architected production RAG-based AI agent platform, demonstrating enterprise AI integration in data operations (automated alert triage, log analysis, intelligent data discovery)
- Developed custom API automation modules enabling AI agents to query platform metadata, event types, and access roles via RESTful endpoints
- Led LLM-driven workflow automation for AI-assisted alert triage, log analysis, and intelligent pipeline monitoring
- Shared DevOps best practices through documentation, code reviews, and knowledge-sharing sessions on CI/CD, Kubernetes, Helm, and Maven workflows
- Collaborated on Snowflake historical data extraction and cross-team technical initiatives

### RF CUNY (Research Foundation of CUNY) — Software Engineer

New York City Metropolitan Area | 2020 – 2022

- Developed full-stack web applications and RESTful APIs with modern frontend frameworks, responsive design, and intuitive UX

- Developed web applications and backend services using Java, Python, and JavaScript, following best practices for code quality, testing, and maintainability
- Built backend services handling authentication, data persistence, and third-party API integrations
- Designed and deployed production websites with a focus on performance, accessibility, and SEO (Google Analytics, conversion tracking)
- Built and maintained CI/CD pipelines for automated testing, code quality checks, and deployment to cloud platforms
- Gained foundational expertise in software architecture, relational databases, and agile development methodologies

EDUCATION

Arizona State University (ASU)

Master's Degree — M.S. Computer Science 2022 – Present

City University of New York (CUNY)

Bachelor's Degree — B.S. Computer Science & Information Security 2016 – 2022

**Relevant Coursework:** Data Processing at Scale, Discrete Structures, Advanced Programming Techniques (*Honors*), Data Structures (*Honors*), Cryptography and Crypto Analysis, Applied Cryptography, Computer Systems and Networks, Software Engineering, Database Systems, Operating Systems, Computer Architecture, Information Security Principles

TECHNICAL SKILLS

| Category                   | Technologies                                                                                                      |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| Languages                  | Java 17+, Python, JavaScript/Node.js, Bash/Shell, PowerShell, SQL                                                 |
| Big Data & Processing      | Apache Spark, Apache Storm, Apache Airflow, Apache Kafka, Apache Iceberg, Nessie, Vertica, Snowflake, Apache Kudu |
| Search & Metadata          | Elasticsearch, Painless Scripting, Elastic SQL                                                                    |
| Cloud & Infrastructure     | Microsoft Azure, Azure Kubernetes Service (AKS), Helm, Docker, ZooKeeper                                          |
| DevOps & CI/CD             | GitLab CI/CD, SonarQube, Maven, JaCoCo, Docker Image Pipelines, Automated Change Management                       |
| APIs & Microservices       | RESTful APIs, JWT/OAuth, Spring WebFlux, Microservice Architecture                                                |
| Frontend Frameworks        | React, React Native, Angular, Responsive Design, UI/UX                                                            |
| Monitoring & Observability | Splunk, Elastic Stack, Spark History Server                                                                       |
| AI/ML & Automation         | AI Agent Development, RAG, Prompt Engineering, LLM Integration, Workflow Automation                               |
| Security & Governance      | SailPoint, Vulnerability Scanning, Threat Modeling, Kubernetes RBAC, SoD Compliance                               |
| Databases                  | PostgreSQL, Vertica, Snowflake, SQL                                                                               |
| Other                      | React Native, IBM Blockchain, Google Analytics, Jira, Git, Insomnia/Postman                                       |

LANGUAGES

- **English** — Native / Bilingual Proficiency
- **Arabic** — Native / Bilingual Proficiency